

Ex09. Introduction to Signals and systems

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1. Plot FIR Filter Signals

This assignment involves generating six figures for an FIR filter using MATLAB functions such as meshgrid, plot3, stem, meshc, roots, zplane, and phase. For the numerical computations, the constant a is set to 0.5, with z defined as ω . The resulting waveforms are organized into a 3 * 2 subplot grid to facilitate comparative analysis.

```
a = 0.5;
r = 0:0.01:1;
omega = -pi:0.01:pi;
[R, OMEG] = meshgrid(r, omega);
Z = R .* exp(1j * OMEG);
n = -10:1:10;

h = @(n) a.^n .* ((n >= 0) - (n >= 7));
H = @(z) ((1 - a.^7 .* z.^(-7)) ./ (1 - a .* z.^(-1)));

figure('Position', [100, 100, 600, 800]);

subplot(3, 2, 1);
p1 = stem(n, h(n), 's-');
title('1. stem Impulse response h[n]');
xlabel('n');
ylabel('h[n]');
grid on;

subplot(3, 2, 2);
b = a.(0:6);
p2 = zplane(b, 1);
title('Zeros and Poles of H(z)');
xlabel('Real');
ylabel('Imaginary');
grid on;

subplot(3, 2, 4);
p4 = meshc(real(Z), imag(Z), 20 * log10(abs(H(Z))));
title('4. meshc Transfer function H(z)');
xlabel('Real(z)');
ylabel('Imag(z)');
zlabel('H(z)');
grid on;

subplot(3, 2, 6);
p6 = plot3(real(Z), imag(Z), 20 * log10(abs(H(Z))));
title('6. plot3 Transfer function H(z)');
```

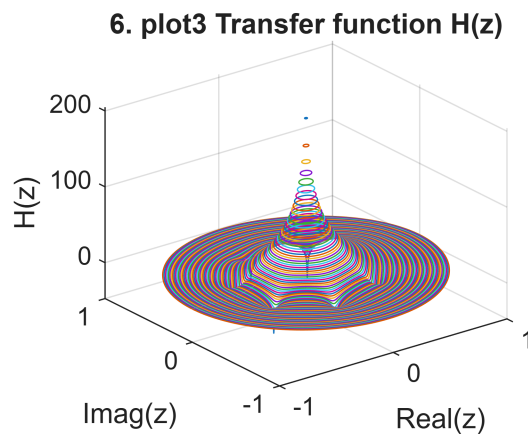
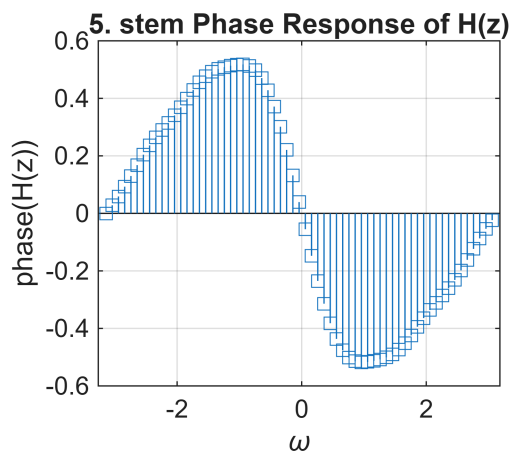
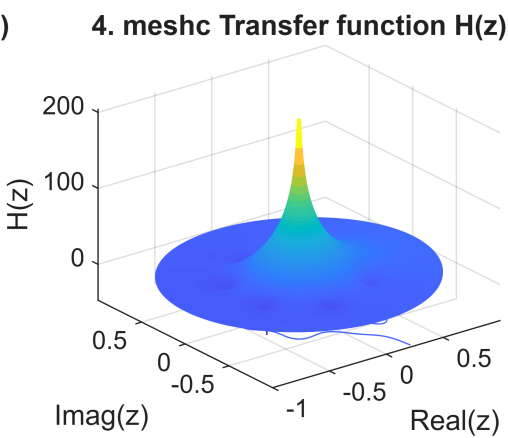
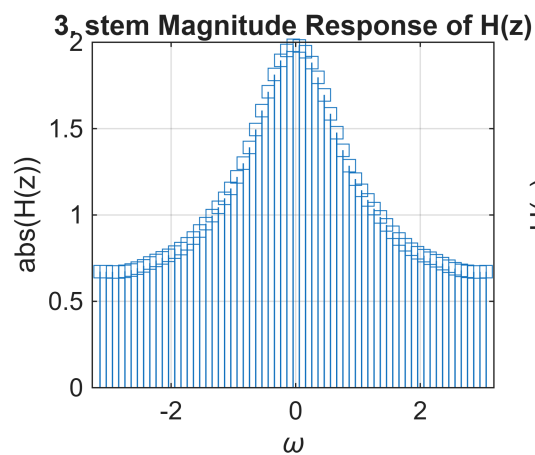
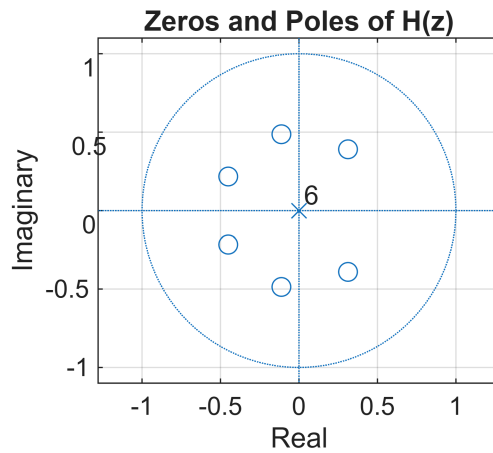
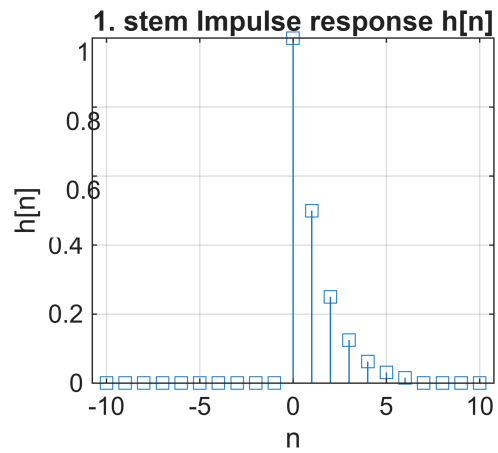
```

xlabel('Real(z)');
ylabel('Imag(z)');
zlabel('H(z)');
grid on;

subplot(3, 2, 3);
r = 1;
omega = -pi:0.1:pi;
[R, OMEG] = meshgrid(r, omega);
Z = r .* exp(1j * OMEG);
p3 = stem(OMEG, abs(H(Z)), 's-');
title('3. stem Magnitude Response of H(z)');
xlabel('\omega');
ylabel('abs(H(z))');
grid on;

subplot(3, 2, 5);
p5 = stem(OMEG, phase(H(Z)), 's-');
title('5. stem Phase Response of H(z)');
xlabel('\omega');
ylabel('phase(H(z))');
grid on;

```



2. Math Problem

Below is the snapshot of my answer in fill-blank math problem.

Ex 9

Chapter 9. Structures for Digital Filters I: FIR Filter

Date (yy/mm/dd): 2026 / 5 / 8

Class: [CSI] [UIT-SPR] ID: s1320112

Name: Motokisa Yamano

Score: ____/100

Math Problem: (4 × 10 points)

- Given an impulse response of a FIR filter is

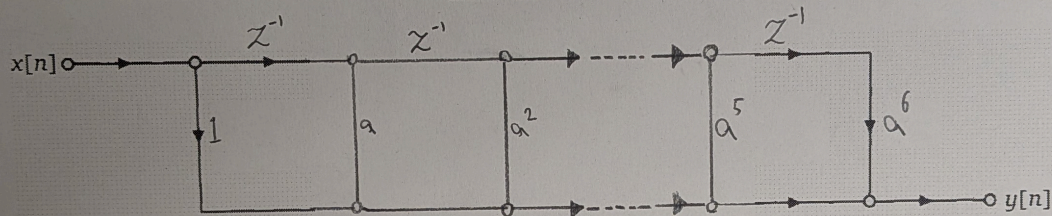
$$h[n] = \begin{cases} \alpha^n, & 0 \leq n \leq 6 \\ 0, & \text{otherwise} \end{cases}$$

- Sketch the direct form implementation of this filter;
- Validate that the corresponding transfer function is

$$H(z) = \frac{1 - \alpha^7 z^{-7}}{1 - \alpha z^{-1}}$$

Answer:

- With an impulse response $h[n] = \alpha^n [u[n] - u[n-7]]$, the direct form implementation of this filter is:



- The transfer function is:

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] z^{-n} = \sum_{n=0}^6 \alpha^n z^{-n} = 1 + \frac{\alpha}{z} + \left(\frac{\alpha}{z}\right)^2 + \dots + \left(\frac{\alpha}{z}\right)^6 = \frac{1 - \alpha^7 z^{-7}}{1 - \alpha z^{-1}}$$

MATLAB Problem: (6 × 10 points)

- Develop a MATLAB program to do the following tasks and submit your results including 6 charts and the MATLAB .m file. ($\alpha = 0.5$; $z = re^{j\omega}$)

Hint: please try to use some commands such as meshgrid, plot3, stem, meshc, roots, zplane, and phase.

- Plot the above unit sample response of an FIR filter $h[n]$ in subplot 321 by using `stem(n, h, 's-')`; ($n = -10:1:10$);
- Calculate $H(z)$ for $r = 0:0.01:1$; $\omega = -\pi:0.01:\pi$. `meshc(real(z), imag(z), 20*log10(abs(H)))` in subplot 324;
- Calculate $H(z)$ for $r = 0:0.01:1$; $\omega = -\pi:0.01:\pi$. `plot3(real(z), imag(z), 20*log10(abs(H)))` in subplot 326;
- Use commands like `roots` and `zplane` to plot zeros/poles placements in subplot 322;
- Plot the magnitude response of $H(z)$ in subplot 323 by using `stem(w, abs(H(z)), 's-')`; ($r = 1$; $\omega = -\pi:0.1:\pi$);
- Plot the phase response of $H(z)$ in subplot 325 by using `stem(w, phase(H(z)), 's-')`; ($r = 1$; $\omega = -\pi:0.1:\pi$);